

Project Checkpoint C: Research Report

Stable Growth Consumer Staples ETF

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Introduction

The third stage of this project aims to develop an optimal trading algorithm for the equities that make up the Stable Growth Consumer Staples ETF (SGCS). This is determined through developing several models and comparing their performance metrics (alpha, beta, Sharpe, etc.) against a baseline.

Literature Review

[1] I used chapter 12 of the Advances in Financial Machine Learning book to get a better understanding of walk through back testing.

Methods

After testing the performance of random equity allocation in “Project Checkpoint B”, the best path forward was to select a few trading strategies and compare them to the null hypothesis. I started by splitting the historical pricing data into training and testing datasets where the training set is price data from January 1, 2000, through December 31, 2019, and the testing set is price data from January 1, 2020, through December 31, 2024. Splitting the data into training and testing allows me to prevent data leakage that may impact the performance of my model when determining its performance.

I decided to test two different trading algorithms on my ETF: momentum trading and mean-price reversion. For both of these strategies, I created a set of parameters to search through to find the optimal version of each type of strategy. For example, mean-reversion is a strategy that assumes equity prices will eventually revert to the mean. Therefore, the optimal strategy is to buy when the price is below the mean and sell when the price is above the mean. One parameter that can be used to tune this strategy is the mean lookback period which determines how far back the algorithm should look to determine the mean price. The rest of the parameters for both momentum trading and mean reversion are as follows:

Mean-Reversion Parameters:

Lookback Period: How far back the algorithm should look to determine the mean price.

Quantiles: How many equities should be tradable in each trading period.

Long Short: Whether taking short positions should be introduced.

Momentum Parameters:

Lookback Period: How far back the algorithm should look to determine price momentum.

Long Short: Whether taking short positions should be introduced.

Neutral Band: The price band through which trading should be halted.

After searching for the optimal hyperparameters for each of these strategies, I took the top five strategies by Sharpe ratio along with the benchmark (\$VDC) and compared their CAGR, Sharpe, volatility, maximum drawdown, and the end value of the investment [2, 3]. Following this comparison, I took the top trading strategy and passed it through a Monte Carlo simulation to determine how this strategy may perform in the future compared to \$VDC [4, 5]. I also performed a walk-forward back test to cross-validate the back testing results [6]. Lastly, I calculated the Alpha and Beta for this strategy comparing the walk-forward back test against the performance of \$VDC over the same period [7].

Conclusions

Through the initial investigation of best trading strategies when using the train-test split of 2000-2019 and 2020-2024, the top performing model by Sharpe Ratio was mean-reversion with a 10-day lookback period, trading 40% of the equities per trading period, and taking only long positions. These configurations resulted in a performance of 13% CAGR, 0.7 Sharpe Ratio, .21 volatility, and a maximum drawdown of 30%. Compared to the baseline of \$VDC, the CAGR is +5% higher and the Sharpe Ratio is +.16, but the volatility is .05 higher and the maximum drawdown is 0.05 greater as well [2, 3].

To continue to assess whether this strategy is viable, I built a Monte Carlo simulation of both \$VDC and the SGCS ETF to simulate performance over the next 5 years. This resulted in significant outperformance compared to the baseline, with returns almost doubled although a slightly higher volatility [4, 5]. The walk-forward back test also produced favorable results showing a consistently high CAGR and Sharpe Ratio over a 20-year period with a relatively low

drawdown [6]. Lastly, I calculated the Alpha and Beta of the strategy over \$VDC from 2020 through 2024 and the strategy produces an alpha of 9% annualized with a beta of 0.7 [7]. These results clearly show that the strategy outperforms the baseline considerably and should be considered for investment.

References

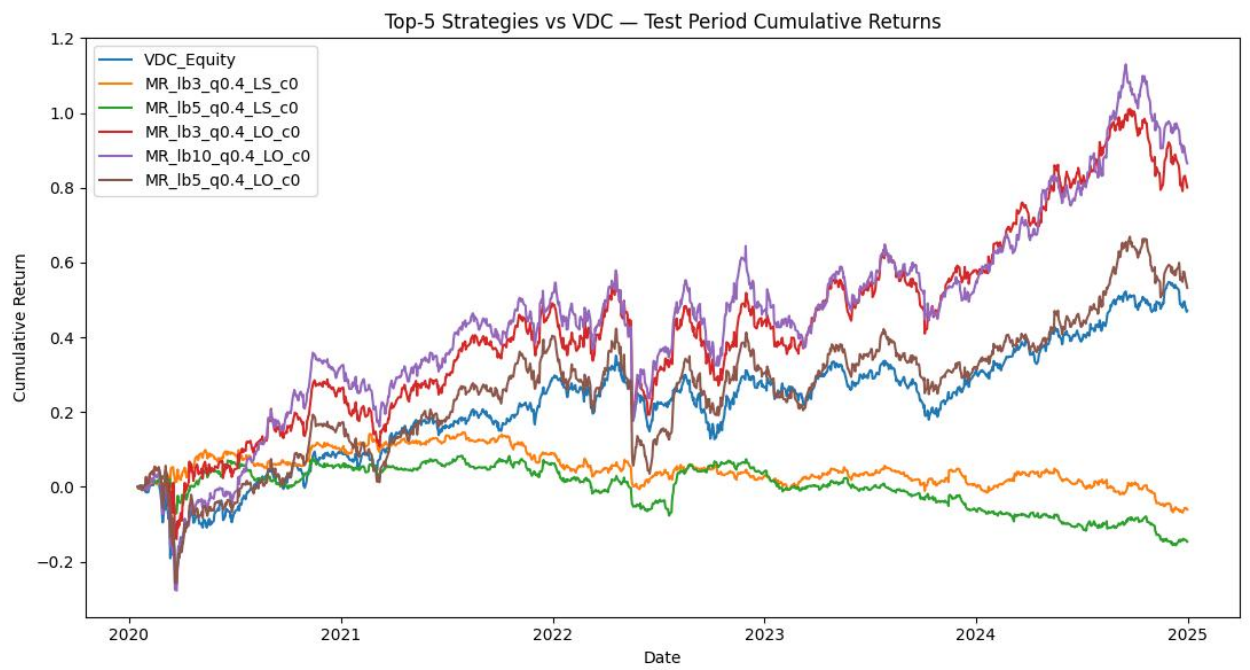
[1] Lopez de Prado, Marcos. “Chapter 12 - Backtesting through Cross-Validation.” Essay. In *Advances in Financial Machine Learning*, 1st ed. Wiley, n.d.

[2] Figure 1

=== Test Summary (Top-5 vs VDC) ===
shape: (6, 6)

Series	CAGR	Sharpe	Vol	MaxDD	EndValue
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str	f64	f64	f64	f64	f64
MR_lb10_q0.4_LO_c0	0.134146	0.703087	0.211161	-0.306924	1.865273
MR_lb3_q0.4_LO_c0	0.126083	0.68183	0.205255	-0.242262	1.800515
VDC_Equity	0.081073	0.544875	0.169652	-0.253069	1.471167
MR_lb5_q0.4_LO_c0	0.089991	0.524529	0.204691	-0.297765	1.53226
MR_lb3_q0.4_LS_c0	-0.012576	-0.117612	0.080305	-0.187459	0.939246
MR_lb5_q0.4_LS_c0	-0.031511	-0.338234	0.08425	-0.221109	0.853366

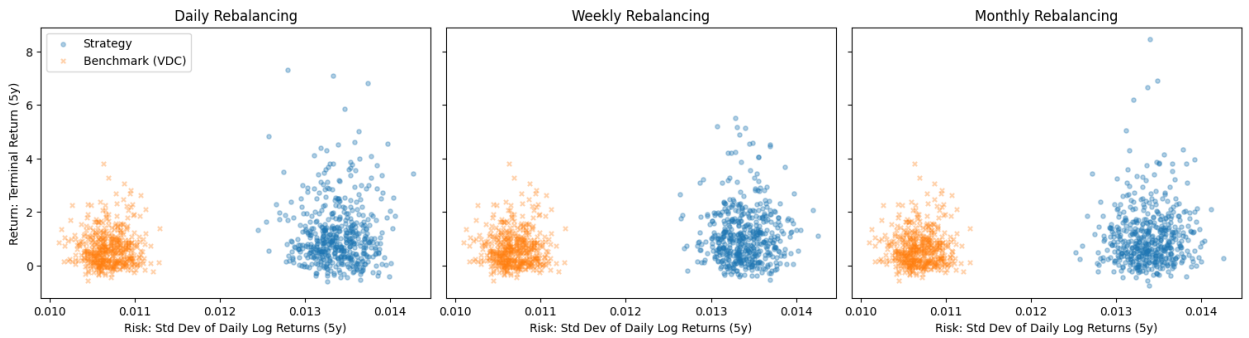
[3] Figure 2



[4] Figure 3

	MeanTerminal	StdTerminal	MeanReturn(%)	Volatility(Daily)	AvgMaxDD(%)	VaR_5%	CVaR_5%
Rebalance							
Benchmark (VDC)	\$1,630,193	\$642,968	63.02	0.0107	-27.94	\$848,987	\$741,001
Daily (Strategy)	\$2,151,771	\$1,153,851	115.18	0.0134	-31.49	\$841,134	\$680,965
Weekly (Strategy)	\$2,108,856	\$1,021,805	110.89	0.0134	-31.60	\$882,472	\$728,818
Monthly (Strategy)	\$2,060,719	\$1,078,864	106.07	0.0134	-31.83	\$875,414	\$699,553

[5] Figure 4



[6] Figure 5

	TrainStartYear	TrainEndYear	TestYear	Strategy	MR_lookback	MR_quantile	MR_long_short	MOM_lookback	MOM_long_short	MOM_neutral_band	trans_cost_bps	Test_CAGR	Test_Sharpe	Test_Volatility	Test_Max Drawdown	Test_Win Rate	Test_Avg Daily Turnover
0	2000	2004	2005	MR	3	0.400000	True	None	None	None	0.000000	3.95%	0.74	5.46%	-2.92%	52.21%	38.44%
1	2001	2005	2006	MR	5	0.400000	True	None	None	None	0.000000	2.46%	0.46	5.59%	-3.71%	48.37%	29.39%
2	2002	2006	2007	MR	10	0.400000	True	None	None	None	0.000000	12.02%	2.01	5.72%	-2.69%	55.42%	22.04%
3	2003	2007	2008	MR	10	0.400000	False	None	None	None	0.000000	-3.15%	0.02	27.88%	-20.05%	55.37%	24.76%
4	2004	2008	2009	MR	5	0.400000	True	None	None	None	0.000000	11.38%	1.27	8.79%	-5.18%	52.85%	30.07%
5	2005	2009	2010	MR	5	0.200000	True	None	None	None	0.000000	15.63%	1.94	7.65%	-5.75%	52.44%	42.45%
6	2006	2010	2011	MR	5	0.200000	True	None	None	None	0.000000	-4.42%	-0.45	9.05%	-8.26%	49.80%	39.43%
7	2007	2011	2012	MR	5	0.200000	True	None	None	None	0.000000	-11.88%	-1.41	8.69%	-14.53%	42.62%	34.36%
8	2008	2012	2013	MR	5	0.200000	True	None	None	None	0.000000	11.82%	1.48	7.74%	-5.25%	56.10%	44.40%
9	2009	2013	2014	MR	10	0.200000	False	None	None	None	0.000000	27.00%	2.15	11.44%	-7.21%	66.39%	32.08%
10	2010	2014	2015	MR	10	0.200000	False	None	None	None	0.000000	-5.53%	-0.33	14.12%	-16.60%	53.11%	30.83%
11	2011	2015	2016	MR	5	0.400000	False	None	None	None	0.000000	19.73%	1.76	10.55%	-4.04%	67.61%	32.52%
12	2012	2016	2017	MR	5	0.400000	False	None	None	None	0.000000	13.58%	1.49	8.78%	-8.36%	66.26%	30.61%
13	2013	2017	2018	MR	3	0.200000	False	None	None	None	0.000000	7.77%	0.50	18.04%	-12.45%	58.70%	48.78%
14	2014	2018	2019	MR	10	0.400000	False	None	None	None	0.000000	35.67%	3.31	9.35%	-5.64%	70.12%	26.39%
15	2015	2019	2020	MR	3	0.400000	False	None	None	None	0.000000	28.09%	1.06	26.58%	-14.68%	62.65%	38.84%
16	2016	2020	2021	MR	3	0.400000	False	None	None	None	0.000000	13.22%	1.29	9.99%	-7.32%	68.15%	38.33%
17	2017	2021	2022	MR	3	0.400000	False	None	None	None	0.000000	16.57%	1.10	14.94%	-8.78%	62.10%	37.79%
18	2018	2022	2023	MR	3	0.400000	False	None	None	None	0.000000	7.93%	0.79	10.33%	-10.84%	64.78%	37.26%
19	2019	2023	2024	MR	3	0.400000	False	None	None	None	0.000000	15.92%	1.67	9.09%	-7.14%	59.68%	38.87%

[7] Figure 6

	TrainStartYear	TrainEndYear	TestYear	Strategy	MR_lookback	MR_quantile	MR_long_short	MOM_lookback	MOM_long_short	MOM_neutral_band	trans_cost_bps	Test_CAGR	Test_Sharpe	Test_Volatility	Test_Max Drawdown	Test_Win Rate	Test_Avg Daily Turnover	Alpha_Annual	Beta
15	2015	2019	2020	MR	3	0.400000	False	None	None	None	0.000000	28.09%	1.06	26.58%	-14.68%	62.65%	38.84%	17.49%	0.84
16	2016	2020	2021	MR	3	0.400000	False	None	None	None	0.000000	13.22%	1.29	9.99%	-7.32%	68.15%	38.33%	2.07%	0.64
17	2017	2021	2022	MR	3	0.400000	False	None	None	None	0.000000	16.57%	1.10	14.94%	-8.78%	62.10%	37.79%	16.21%	0.63
18	2018	2022	2023	MR	3	0.400000	False	None	None	None	0.000000	7.93%	0.79	10.33%	-10.84%	64.78%	37.26%	6.06%	0.70
19	2019	2023	2024	MR	3	0.400000	False	None	None	None	0.000000	15.92%	1.67	9.09%	-7.14%	59.68%	38.87%	7.33%	0.61